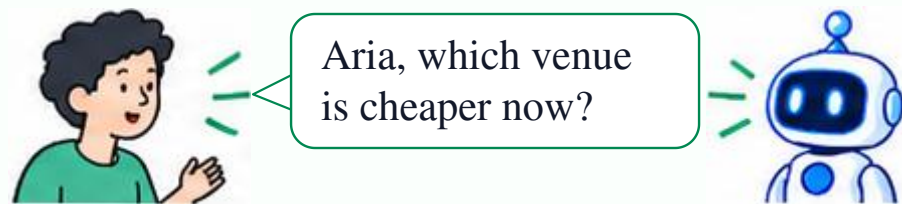




Audio input type in the multi-party room



Direct request to Aria



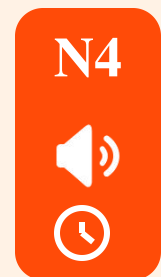
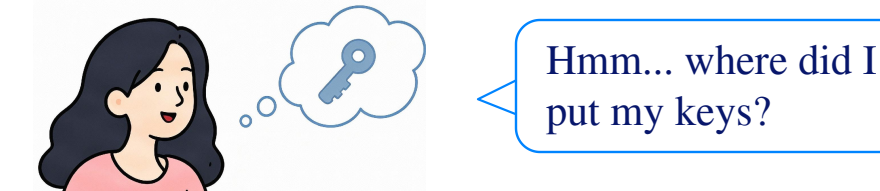
Mention only



Talking to someone else



Self-talk



Resolved request



T = Speak



N1–N3 = Stay silent



N4 = Speak while unresolved
→ stop after resolution



Audio detail to SDMs

Spoken duty preamble

“You are the voice assistant Aria, that is A R I A. Aria. You are in a room with several people...”

A
B
C
B
C
A

Oak quoted \$40 for the room and \$12 per person.

N2

Pine is \$18 per person, with no room fee.

N2

Six of us confirmed, including Maya.

N2

Pine also adds a \$5 cleaning charge.

N2

Maya just cancelled; only five of us can come.

N2

Aria, which venue is cheaper now, and by how much?

T

Continuous multi-party audio

Spoken Dialogue Model

For five people:
Oak: $\$40 + 5 \times \$12 = \$100$.
Pine: $5 \times \$18 + \$5 = \$95$.
Pine is \$5 cheaper.



Benchmark Metrics



Fresh-onset response rate



Conditional answer accuracy



Silence preservation



Answering-window yield



Dialogue state during N4



Human asker
Aria (assistant)

Human resolver

Aria, when does the store close?

The store closes at around...

I just checked — it closes at nine.

Aria should keep silent